```
import time
import pandas as pd

from sklearn.datasets import fetch_california_housing
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error
```

```
data=fetch_california_housing()
print(data.keys())
```

```
dict_keys(['data', 'target', 'frame', 'target_names', 'feature_names', 'DESCR'])
```

```
x=pd.DataFrame(data.data,columns=data.feature_names)
y=data.target

print(x.head())
print("samples:",len(x))
print(y[:10])
```

```
   MedInc  HouseAge  AveRooms  AveBedrms  Population  AveOccup  Latitude  \
0  8.3252      41.0  6.984127   1.023810       322.0  2.555556     37.88
1  8.3014      21.0  6.238137   0.971880      2401.0  2.109842     37.86
2  7.2574      52.0  8.288136   1.073446       496.0  2.802260     37.85
3  5.6431      52.0  5.817352   1.073059       558.0  2.547945     37.85
4  3.8462      52.0  6.281853   1.081081       565.0  2.181467     37.85

   Longitude
0    -122.23
1    -122.22
2    -122.24
3    -122.25
4    -122.25
samples: 20640
[4.526 3.585 3.521 3.413 3.422 2.697 2.992 2.414 2.267 2.611]
```

```
x_train,x_test,y_train,y_test=train_test_split(
    x,y,test_size=0.2,random_state=42
)
```

```
tree=DecisionTreeRegressor()

start=time.perf_counter()
tree.fit(x_train,y_train)
tree_time=time.perf_counter() - start

pred_tree_test=tree.predict(x_test)
err_tree_test=mean_absolute_error(y_test,pred_tree_test)

print("decision tree train mae:",err_tree_test)

pred_tree_train=tree.predict(x_train)
err_tree_train=mean_absolute_error(y_train,pred_tree_train)

print("decision tree train mae:",err_tree_train)
print("decision tree test mae:",tree_time)
```

```
decision tree train mae: 0.4520883212209303
decision tree train mae: 4.125683429687476e-17
decision tree test mae: 0.37336710400006723
```

```
rforest=RandomForestRegressor()

start=time.perf_counter()
rforest.fit(x_train,y_train)
rforest_time=time.perf_counter() - start

pred_rforest_test=rforest.predict(x_test)
err_rforest_test=mean_absolute_error(y_test,pred_rforest_test)
```

```
pred_rforest_train=rforest.predict(x_train)
err_rforest_train=mean_absolute_error(y_train,pred_rforest_train)

print(" rforest train mae:",err_rforest_train)
print(" rforest test mae:",err_rforest_test)
print(" rforest time:",rforest_time)
```

```
 rforest train mae: 0.12230059872819805
 rforest test mae: 0.32791615237403116
 rforest time: 43.260862342999985
```

```
gb=GradientBoostingRegressor(n_estimators=300, learning_rate=0.1, max_depth=5)

start=time.perf_counter()
gb.fit(x_train,y_train)
gb_time=time.perf_counter() - start

pred_gb_test=gb.predict(x_test)
err_gb_test=mean_absolute_error(y_test,pred_gb_test)

pred_gb_train=rforest.predict(x_train)
err_gb_train=mean_absolute_error(y_train,pred_gb_train)

print("gb train mae:",err_gb_train)
print("gb test mae:",err_gb_test)
print("gb time:",gb_time)
```

```
gb train mae: 0.12230059872819805
gb test mae: 0.30698970727711666
gb time: 29.769354418000148
```

```
print("\n ==== Comparison For Test Mae ==== \n")
print("decision_tree_test:",err_tree_test)
print("random_forest_test:",err_rforest_test)
print("gradient_boost_test:",err_gb_test)
```

```
 ==== Comparison For Test Mae ====

decision_tree_test: 0.4520883212209303
random_forest_test: 0.32791615237403116
gradient_boost_test: 0.30698970727711666
```

Start coding or generate with AI.